

Is the Jackpot a Lie ?
A Data-Driven Exploration of Lottery Randomness

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Audience

This analysis is designed for general public who frequently participate in lottery games though may not have formal statistical training. The goal is to share insights clearly without technical jargon.

Purpose

The purpose of this is to find out if lottery results have patterns which could indicate manipulation or if they are simply a result of statistical randomness. The call to action is to encourage players to make educated decisions and understand that perceived patterns are often cognitive bias rather than evidence of rigging.

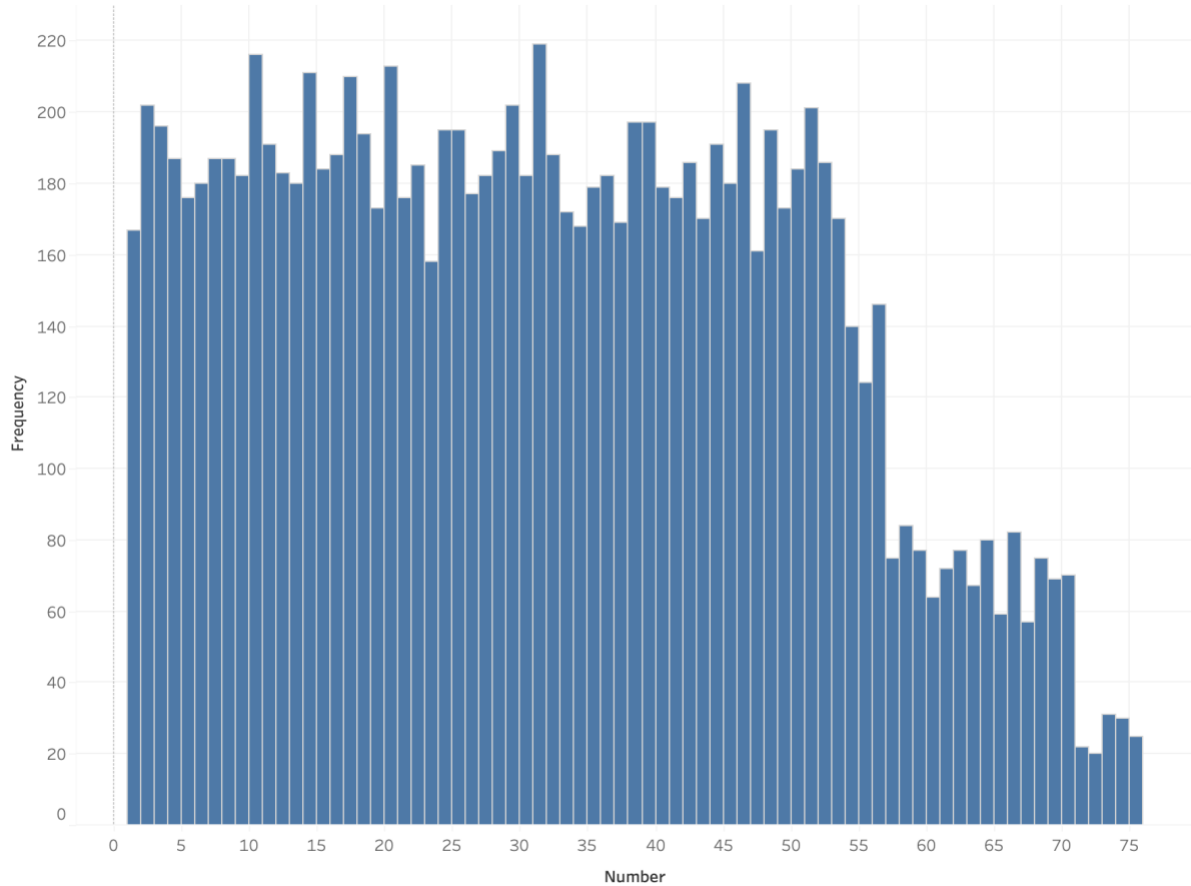
Medium

The visuals were created via a Tableau dashboard and exported to powerpoint. I prepared the summary in Microsoft Word, including the exported visuals to support the narrative explanation and final submission was done as a PDF document. This approach allows structured storytelling by combining visual analysis with written interpretation.

Visuals and Findings

Visual 1: Mega Millions Histogram

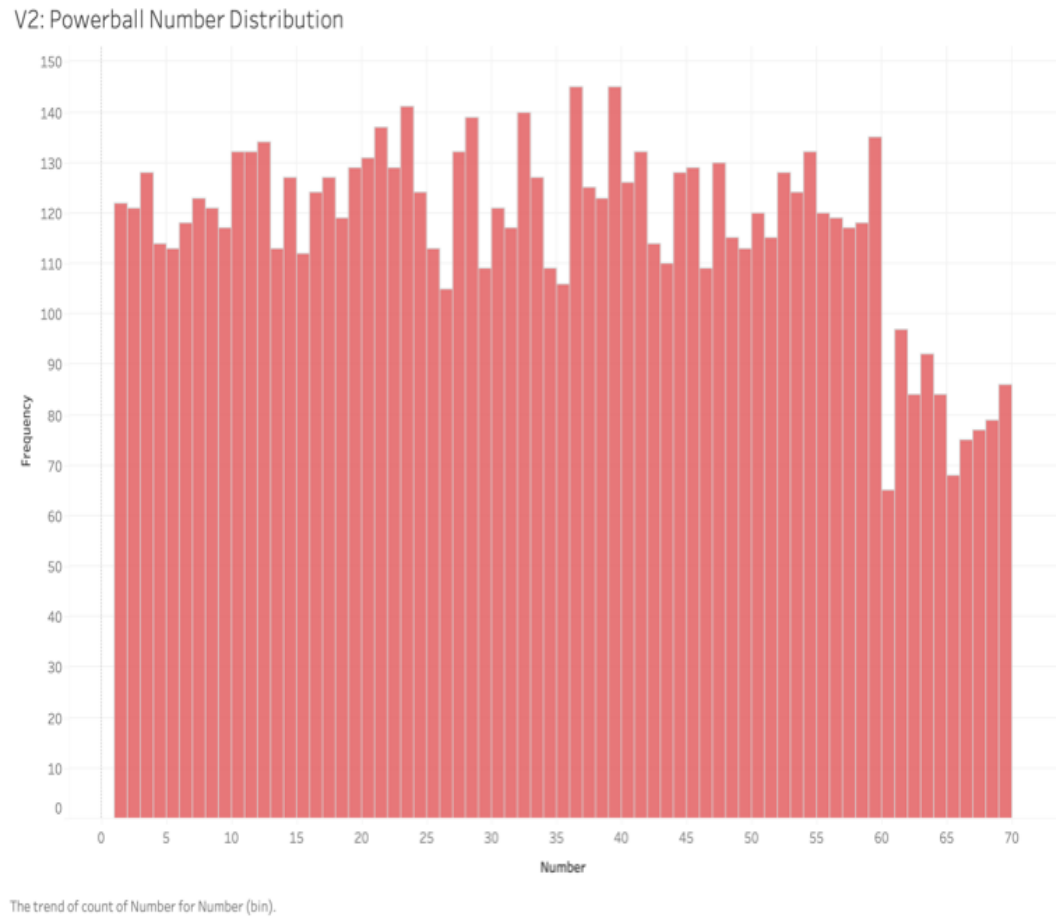
V1: Mega Millions Number Distribution



The trend of count of Number for Number (bin).

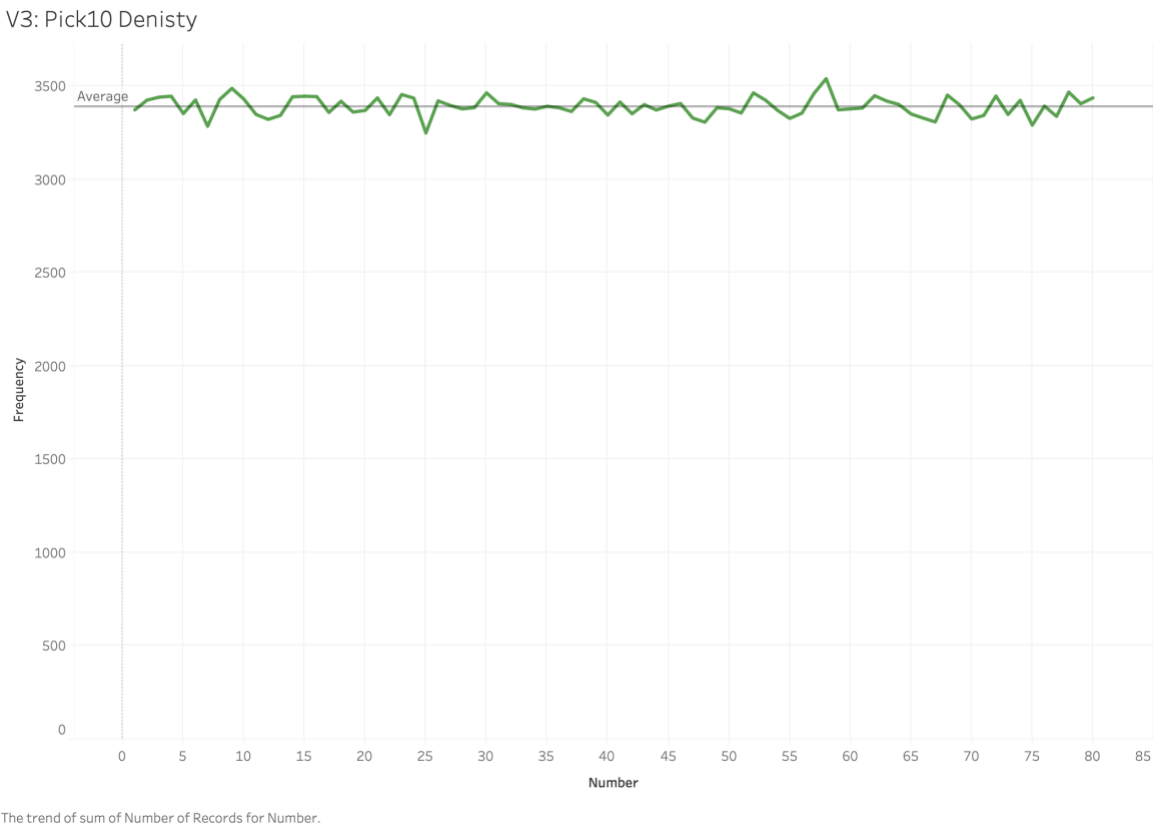
This histogram represents the frequency distribution of winning numbers compared to the overall range of numbers. The bars seem relatively balanced, with only slight variations from one number to another.

Visual 2: Powerball Histogram



The Powerball histogram also shows a similar distribution pattern as Mega Millions histogram. There are some numbers that have slightly higher occurrences than others, the distribution pattern is similar for all numbers. There are no gaps or peaks that could indicate any form of bias.

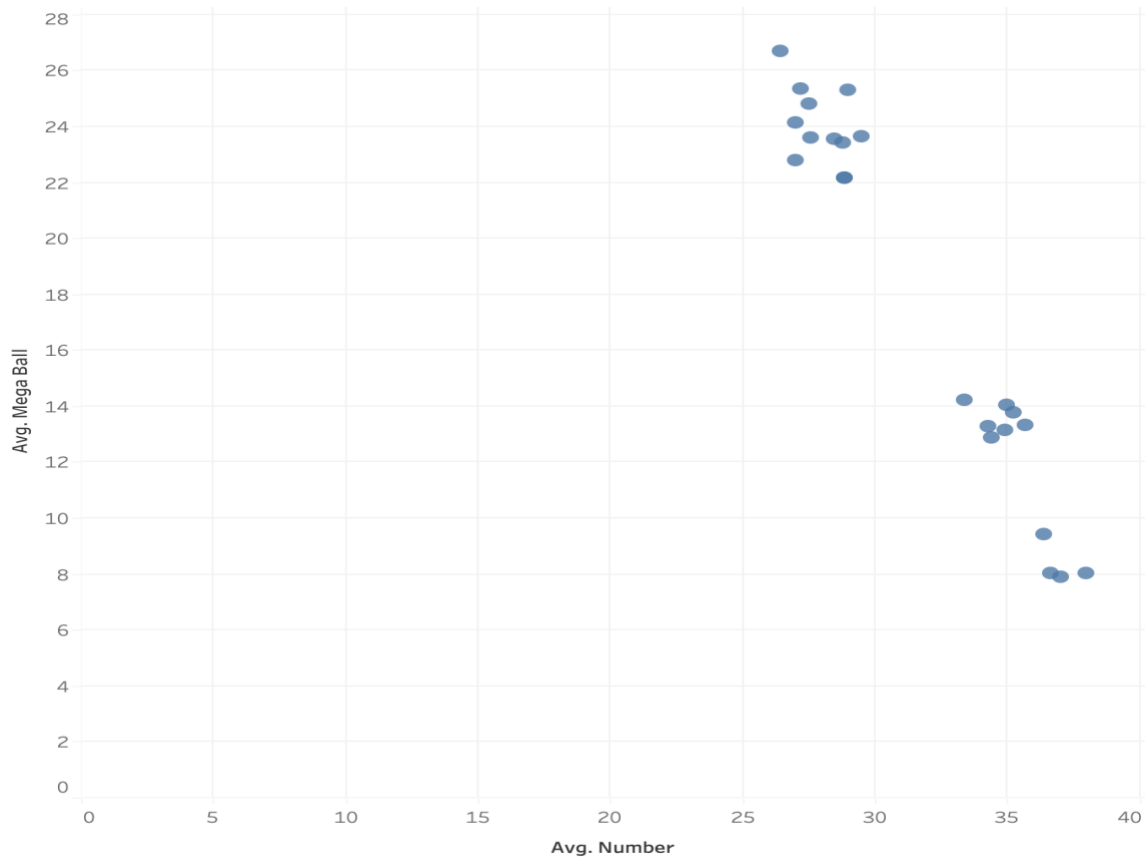
Visual 3: Pick 10 Density Plot



The Pick10 density plot shows the distribution pattern of the picked numbers. The graph is a smooth and uniform pattern typical of random numbers over time.

Visual 4: Mega Millions Scatterplot

V4: Mega Scatterplot

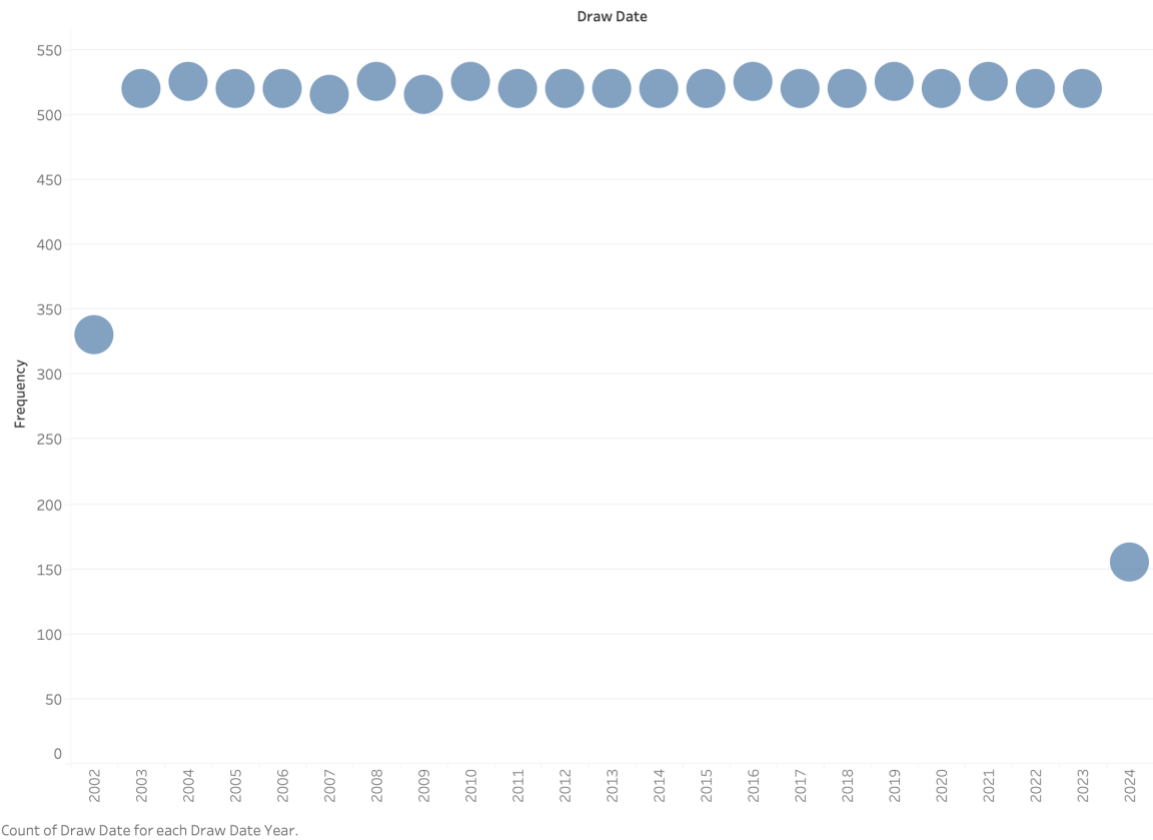


Average of Number vs. average of Mega Ball. Details are shown for Draw Date Year.

The scatter plot shows the main numbers and the Mega Ball numbers. The data points are scattered randomly without any pattern of forming a shape. This indicates that the main numbers and the Mega Ball number are independent of each other. This is an important characteristic of a fair lottery system.

Visual 5: Mega Millions Bubble Chart

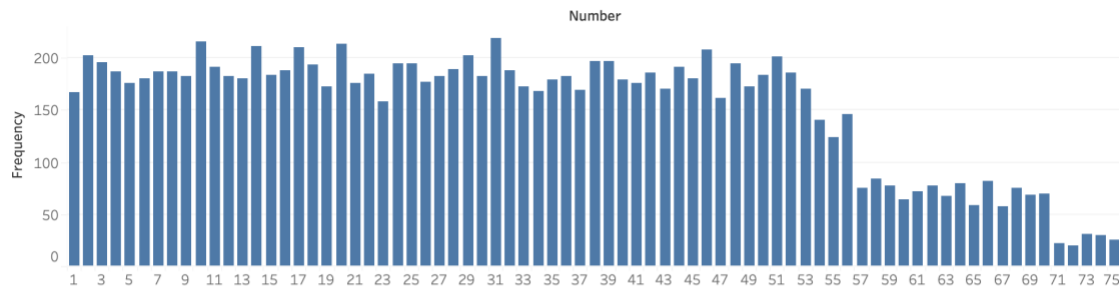
V5: Mega Bubble



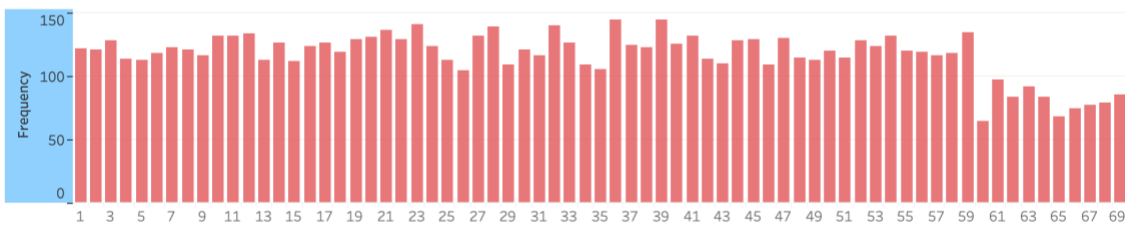
The bubble chart shows the draws per year. The size of the bubbles is constant in most years which show stability.

Visual 6: Small Multiples Comparison (Mega, Powerball, Pick 10)

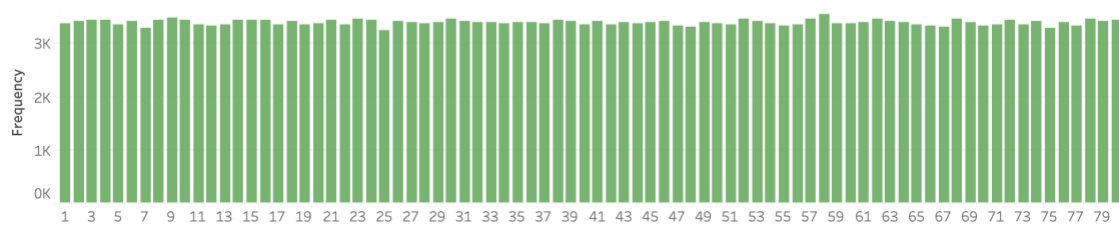
Mega Millions – Number Frequency Distribution



Powerball – Number Frequency Distribution



Pick 10 – Number Frequency Distribution



This small multiples design allows for a comparison of the three games. Each game shows similar patterns of structural distribution.

Design and Gestalt Principles

Similarity and continuity was confirmed by maintaining consistency in the design via uniform bin sizes, aligned axes. Color contrast was applied to differentiate games. Spacing & layout was used based on proximity principles to group related graphics.

Ethical Considerations

All data was taken from the data.ny.gov (public database). The original values were not changed but winning number strings were splitted into separate numeric fields to allow proper analysis and visualization in Tableau. Only simple calculated fields and functions were utilized. No data was deleted. Scales and binning were standardized to prevent

deceptive visualization. All assumptions was restricted to mathematical aggregation, as in probability analysis.

Conclusion

The findings do not support the claim that “the jackpot is a lie.” Results align with expected statistical randomness. Players should recognize that perceived patterns often reflect cognitive bias rather than systemic manipulation.